Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/07/2019
PI Name:	Wu, Honglu Ph.D.		
Project Title:	Cytogenetic Study of Heavy Ion-Induced Chromosomal Damage in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	07/01/2004	**End Date:**	09/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Correct end date is 9/30/2010 per C. Guidry/JSC (Ed., 5-28-2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gonda, Steve (NASA Johnson Space Center)		
Grant/Contract No.:	NA00405		
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	Energetic heavy ions pose a great health risk to astronauts in extended International Space Station (ISS) and future Mars missions. High-LET (linear energy transfer) heavy ions are particularly effective in causing various biological effects, including cell inactivation, genetic mutations and cancer induction. Most of these biological endpoints are closely related to chromosomal damage, which can be an indicator of health risks. In this project, we will quantitatively analyze chromosomal damage induced by energetic heavy ions in human epithelia and lymphocyte cells, by using advanced cytogenetic techniques including the multi-color fluorescence in situ hybridization (mFISH) and mBAND techniques, which will allow the analysis of chromosomal exchanges involving all human chromosomes and intrachromosomal aberrations, respectively. Human epithelia cells will be cultured in both the conventional two-dimensional (2-D) and tissue-equivalent three-dimensional (3-D) platforms. The exposure of human cells to energetic heavy ions will be		

	conducted at the newly completed NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory, and a gamma radiation source at the Baylor College of Medicine will be used for low-LET exposures. Experimental results of the proposed investigation may provide the knowledge for identifying the complexity of chromosomal aberrations due to space-like radiation and for assessing the radiation risks from such end-points observed in differential cell systems.
Task Description:	Our major objectives are to (1) comparatively assess the frequencies of complex and simple, as well as interchromosomal and intrachromosomal, aberrations induced by low- and high-LET radiation, (2) evaluate the pattern and types of heavy-ion induced chromosomal aberrations between human 2-D epithelia cells, 3-D epithelia cells, and lymphocytes. The differential radiation environment encountered in space may cause deleterious effects in humans, and these effects are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. Before such missions can be undertaken, a much more detailed understanding of these effects is required to allow planning and implementation of protective countermeasures. The proposed research will attempt to addresses critical questions within the NASA critical path roadmap (CPR) relevant to the assessment of human risk factors (Risk numbers 38 of the CPR) during space flight due to the deleterious effects of the space radiation environment. The focus of the proposal is centered specifically on assessing chromosomal damages induced by space-like radiation and evaluating genomic disruption in tissue-like platforms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[NOTE: from Principal Investigator's report initiated in 2013 but not completed; completed by Task Book Editor March 2019] This is the final report of the project. A number of manuscripts were published to report the results obtained in the project period. The results of our mBAND study confirmed the observation of a higher incidence of inversions for high-LET radiation. Detailed analysis of the inversion type revealed that both iron ions and gamma rays induced a low incidence of simple inversions. Half of the inversions observed in the low-LET-irradiated samples were accompanied by other types of intrachromosome aberrations, but few inversions were accompanied by interchromosome aberrations. In contrast, iron ions induced a significant fraction of inversions that involved complex rearrangements of both inter- and intrachromosome exchanges (Hada et al., 2007a--see Bibliography section). In the neutron experiments, comparison of the dose responses between gamma-rays and neutron irradiations for the fraction of cells with damaged chromosome 3 yielded a relative biological effectiveness (RBE) value of 26 +/- 4 for the secondary neutrons. The study also revealed that secondary neutrons of low dose rate induced a higher fraction of intrachromosome exchanges than gamma-rays, but the fractions of inversions observed between these two radiation types were indistinguishable. Similar to the findings after acute radiation exposures, most of the inversions observed in the neutron study were accompanied by other aberrations. The fractions of complex type aberrations and of unrejoined chromosomal breakages were also found to be higher in the neutron-exposed cells than after gamma-rays (Hada et al., 2010--see Bibliography section). For all radiation types and dose rates, the distribution of the breaks in chromosome 3 that participated in either inter- or intrachromosome exchanges were also investigated. The breakpoint distributions for all radiation types and dose rates were found to be different from the predicted distribution based on band width. Detailed analysis of chromosome fragment ends involved in inter- or intrachromosomal exchanges revealed that the distributions of fragment ends participating in interchromosomal exchanges were similar between the two low-LET radiation dose rates and between the two high-LET radiation types, but the distributions were less similar between low- and high-LET radiations. For fragment ends participating in intrachromosomal exchanges, the distributions for all four radiation scenarios were similar, with clusters of breaks found in three regions. Analysis of the locations of the two fragment ends in chromosome 3 that joined to form an intrachromosomal exchange demonstrated that two breaks with a greater genomic separation can be more likely to rejoin than two closer breaks, indicating that chromatin folding can play an important role in the rejoining of chromosome breaks. Comparison of the breakpoint distributions to the distributions of genes indicated that the gene-rich regions do not necessarily contain more breaks. In general, breakpoint distributions depend on whether a chromosome fragment joins with another fragment in the same chromosome or with a fragment from a different chromosome (Hada et al., 2011--see Bibliography section). During space travel, astronauts will be exposed to protons and heavy charged particles. Since the proton flux is high compared to HZE (high energy) particles, on average, it is assumed that a cell will be hit by a proton before it is hit by an HZE ion. Although the effects of individual ion species on human cells have been investigated extensively, little is known about the effects of exposure to mixed beam irradiation. To address this, we exposed human epithelial cells to protons followed by HZE particles and analyzed chromosomal damage using mBAND procedure. Our results indicated that chromosome aberration frequencies from exposures to protons followed by Fe ions did not simply decrease as the interval between the two exposures increased, but peak when the interval was 30 min (Hada et al., 2007b--see Bibliography section). We also studied the spatial and temporal distributions of foci of the phosphorylated form of the histone protein H2AX (gamma-H2AX), which is known to be activated by double-strand breaks after irradiation of human fibroblast cells with high energy silicon (54 keV/mm) and iron (176 keV/mm) ions. The image obtained with the ion path parallel to a monolayer of human fibroblast cells leads to gamma-H2AX aggregates in the shape of streaks stretching over several micrometers in an x/y plane, thus enabling the analysis of the fluorescence distributions along the ion trajectories. Qualitative analyses of these images showed that cells after Fe ion irradiation exhibit more residual gamma-H2AX clusters than Si ions, confirming our chromosome results demonstrating that higher LET radiation produces more unrejoined residual chromosome breaks (Desai et al., 2005--see Bibliography section).
Bibliography Type:	Description: (Last Updated: 07/15/2026)

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